

May 2024
IB MAA HL P1 TZ1 Student A

1. mean = 3

$$(a) \quad \frac{p + 2q + 3(4) + 4(2) + 6(3)}{16} = 3.$$

$$p + 2q + 38 = 48$$

$$\Rightarrow p + 2q = 10 \quad - (1)$$

$$p + q + 4 + 2 + 3 = 16$$

$$p + q = 7 \quad - (2)$$

$$(1) - (2): \quad q = 3$$

$$\Rightarrow p = 4.$$

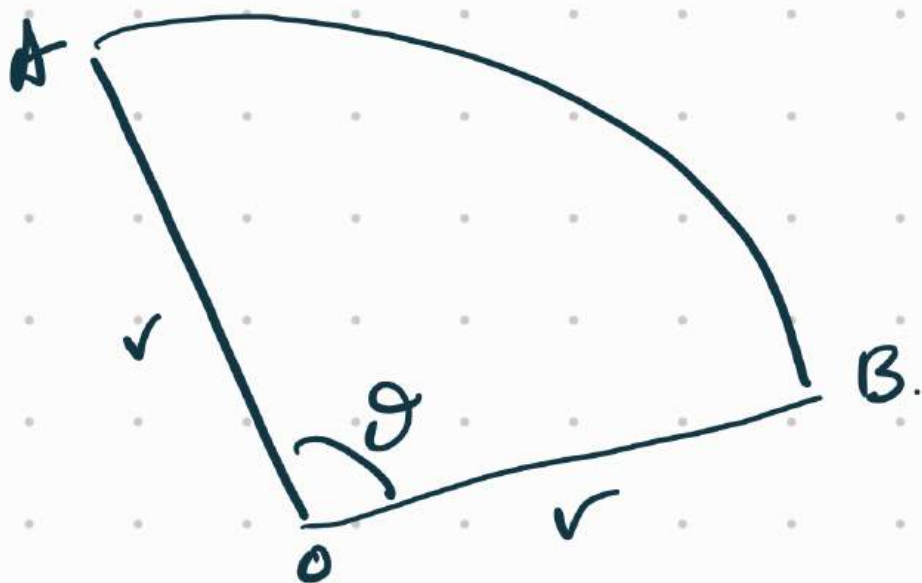
$$(b) \quad \text{mean final score} = 10(3) \\ = 30.$$

$$2. \quad \log_{10} a = \frac{1}{3}, \quad a > 0$$

$$(a) \quad \log_{10} \left(\frac{1}{a} \right) = \log_{10}(1) - \log_{10} a \\ = -\frac{1}{3}.$$

$$(b) \quad \log_{1000} a = \frac{\log_{10} a}{\log_{10} 1000} \\ = \frac{\frac{1}{3}}{\log_{10} 10^3} \\ = \frac{\frac{1}{3}}{3} \\ = \frac{1}{9}$$

3.



$$(a) \text{ Perimeter} = 2r + r\theta = 10 \quad - (1)$$

$$\text{Area} = \frac{1}{2}r^2\theta = 6.25$$

$$\Rightarrow r^2\theta = 12.5$$

$$\Rightarrow \theta = \frac{12.5}{r^2} \quad - (2)$$

sub. (2) into (1).

$$2r + r\left(\frac{12.5}{r^2}\right) = 10$$

$$2r^2 + 12.5 = 10r$$

$$4r^2 - 20r + 25 = 0 \quad (\text{shown})$$

(b)

$$4r^2 - 20r + 25 = 0$$

$$(2r - 5)(2r - 5) = 0$$

$$r = \frac{5}{2}$$

$$\theta = \frac{12 \cdot 5}{r^2} = \frac{12 \cdot 5}{\left(\frac{5}{2}\right)^2}$$

$$= \frac{25}{2} \times \frac{4}{25}$$

$$= 2 \text{ rad.}$$

4. $f(x) = \cos x$, $g(x) = \sin 2x$
 $0 \leq x \leq \pi$.

(a) $\cos x = \sin 2x$.

$\Rightarrow \cos x = 2 \sin x \cos x$

$\cos x (1 - 2 \sin x) = 0$

$\cos x = 0$ or $\sin x = \frac{1}{2}$.

(Rejected
as $\cos x \neq 0$)

$x = \frac{\pi}{6}$ or $\pi - \frac{\pi}{6}$
 $= \frac{5\pi}{6}$.

x -coordinate of $A = \frac{\pi}{6}$

x -coordinate of $C = \frac{5\pi}{6}$

(b) Area of R

$= \int_{\frac{\pi}{2}}^{\frac{5\pi}{6}} (\cos x - \sin 2x) dx$

$= \left[\sin x + \frac{\cos 2x}{2} \right]_{\frac{\pi}{2}}^{\frac{5\pi}{6}}$

$$= \left[\sin\left(\frac{5\pi}{6}\right) + \frac{1}{2} \cos\left(\frac{5\pi}{3}\right) \right] - \left[\sin\frac{\pi}{2} + \frac{1}{2} \cos\pi \right]$$

$$= \left[\frac{1}{2} + \frac{1}{2}\left(\frac{1}{2}\right) \right] - \left[1 - \frac{1}{2} \right]$$

$$= \frac{3}{4} - \frac{1}{2}$$

$$= \frac{1}{4} \text{ units}^2.$$

5. G.P. $u_1 = 1, r = 10.$

$$(a) S_n = \frac{1(10^n - 1)}{10 - 1} = \frac{10^n - 1}{9}$$

$$a = 10, b = 9$$

$$(b) S_1 + S_2 + S_3 + \dots + S_n$$

$$= \frac{10^1 - 1}{9} + \frac{10^2 - 1}{9} + \frac{10^3 - 1}{9} + \dots + \frac{10^n - 1}{9}$$

$$= \frac{1}{9} [10^1 + 10^2 + 10^3 + \dots + 10^n - (1 + 1 + 1 + \dots + 1)]$$

$$= \frac{1}{9} \left[\frac{10(10^n - 1)}{10 - 1} - n \right]$$

$$= \frac{1}{9} \left[\frac{10(10^n - 1) - 9n}{9} \right]$$

$$= \frac{10(10^n - 1) - 9n}{81}$$

6. $f(x) = \sqrt{x \sin(x^2)}, 0 \leq x \leq \sqrt{\pi}$

Volume of revolution about x -axis

$$= \pi \int_0^{\sqrt{\pi}/2} x \sin(x^2) dx$$

$$= \frac{\pi}{2} \int_0^{\sqrt{\pi}/2} 2x \sin(x^2) dx$$

$$= \frac{\pi}{2} \left[-\cos(x^2) \right]_0^{\sqrt{\pi}/2}$$

$$= -\frac{\pi}{2} \left[\cos\left(\frac{\pi}{4}\right) - \cos(0) \right]$$

$$= -\frac{\pi}{2} \left[\frac{\sqrt{2}}{2} - 1 \right]$$

$$= \frac{\pi}{2} \left[\frac{2 - \sqrt{2}}{2} \right]$$

$$= \frac{\pi(2 - \sqrt{2})}{4}$$

7. Prove by Mathematical Induction

$$\sum_{r=1}^n {}^r C_1 = {}^{n+1} C_2, \quad n \in \mathbb{Z}^+.$$

Base Case: $n=1$ LHS = ${}^1 C_1 = 1$

RHS = ${}^2 C_2 = 1 = \text{LHS}$.

True for $n=1$.

Assume that for some $k \in \mathbb{Z}^+$,

$$\sum_{r=1}^k {}^r C_1 = {}^{k+1} C_2.$$

Consider the case

$n = k+1$.

$$\begin{aligned} \text{LHS} &= \sum_{r=1}^{k+1} {}^r C_1 = \sum_{r=1}^k {}^r C_1 + {}^{k+1} C_1 \\ &= {}^{k+1} C_2 + {}^{k+1} C_1 \end{aligned}$$

$$= \frac{(k+1)!}{2! (k+1-2)!} + \frac{(k+1)!}{1! (k+1-1)!}$$

$$= \frac{(k+1)!}{2! (k-1)!} + \frac{(k+1)!}{k!}$$

$$= (k+1)! \left[\frac{k+2}{2! k!} \right]$$

$$= \frac{(k+2)(k+1)!}{2! k!}$$

$$= \frac{(k+2)!}{2! k!} = {}^{k+2}C_2$$

Since it is true for $n=1$, and
true for $n=k \Rightarrow n=k+1$ is true,
by mathematical induction,

$$\sum_{r=1}^n {}^r C_1 = {}^{n+1}C_2, \quad n \in \mathbb{Z}^+$$

8. (a) Maclaurin series:

$$(i) \quad \sin(x^2) = x^2 - \frac{(x^2)^3}{3!} + \dots$$
$$= x^2 - \frac{x^6}{6} + \dots$$

$$(ii) \quad \sin^2(x^2) = \left(x^2 - \frac{x^6}{6} + \dots\right) \left(x^2 - \frac{x^6}{6} + \dots\right)$$
$$= x^4 - \frac{x^8}{6} - \frac{x^8}{6} + \dots$$
$$= x^4 - \frac{x^8}{3} + \dots$$

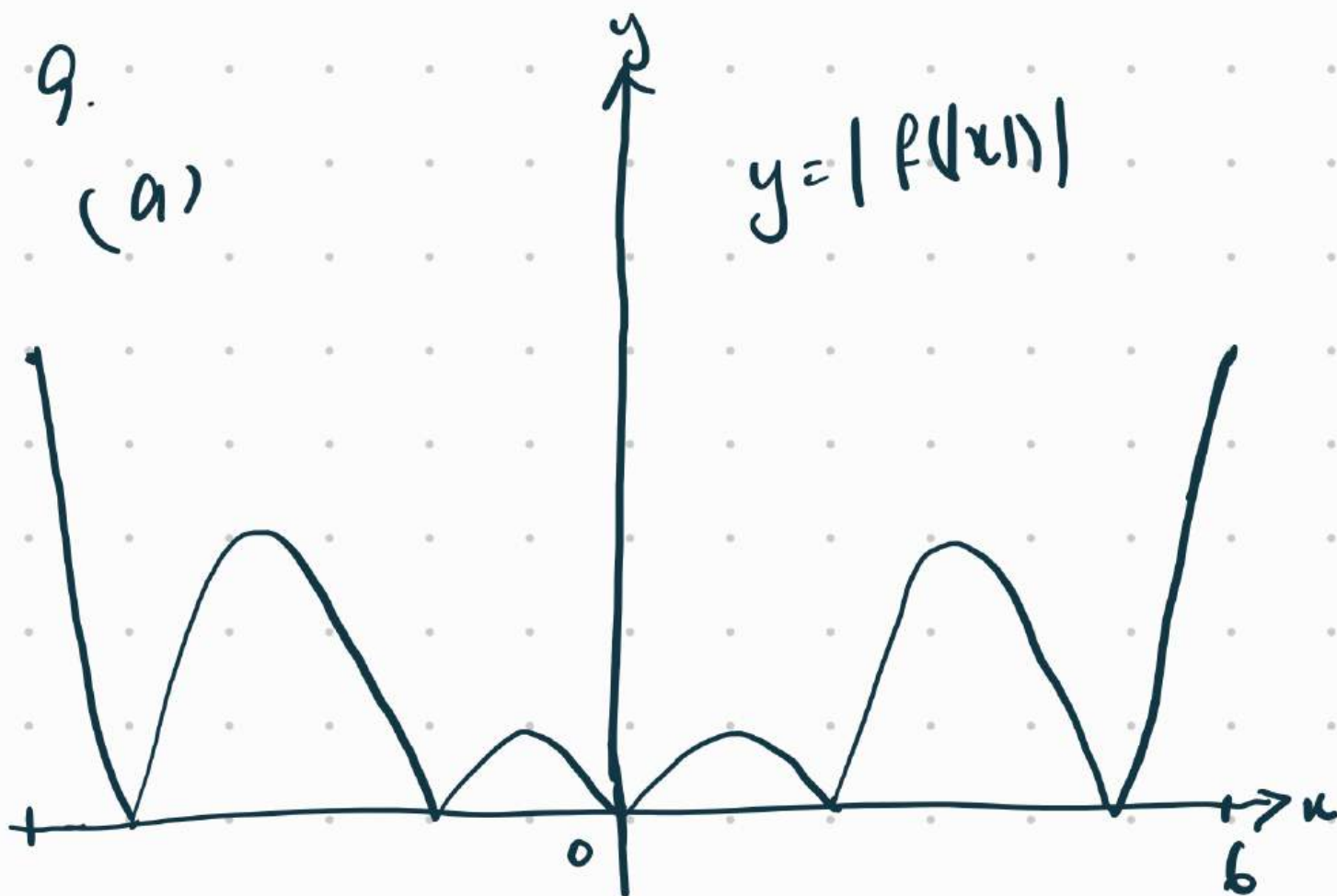
$$(b) \quad 4x \sin(x^2) \cos(x^2)$$
$$= 4x \left(x^2 - \frac{x^6}{6} + \dots\right) \left(1 - \frac{(x^2)^2}{2!} + \dots\right)$$
$$= 4x \left(x^2 - \frac{x^6}{2} - \frac{x^6}{6} + \dots\right)$$
$$= 4x \left(x^2 - \frac{2x^6}{3} + \dots\right)$$
$$= 4x^3 - \frac{8x^7}{3} + \dots$$

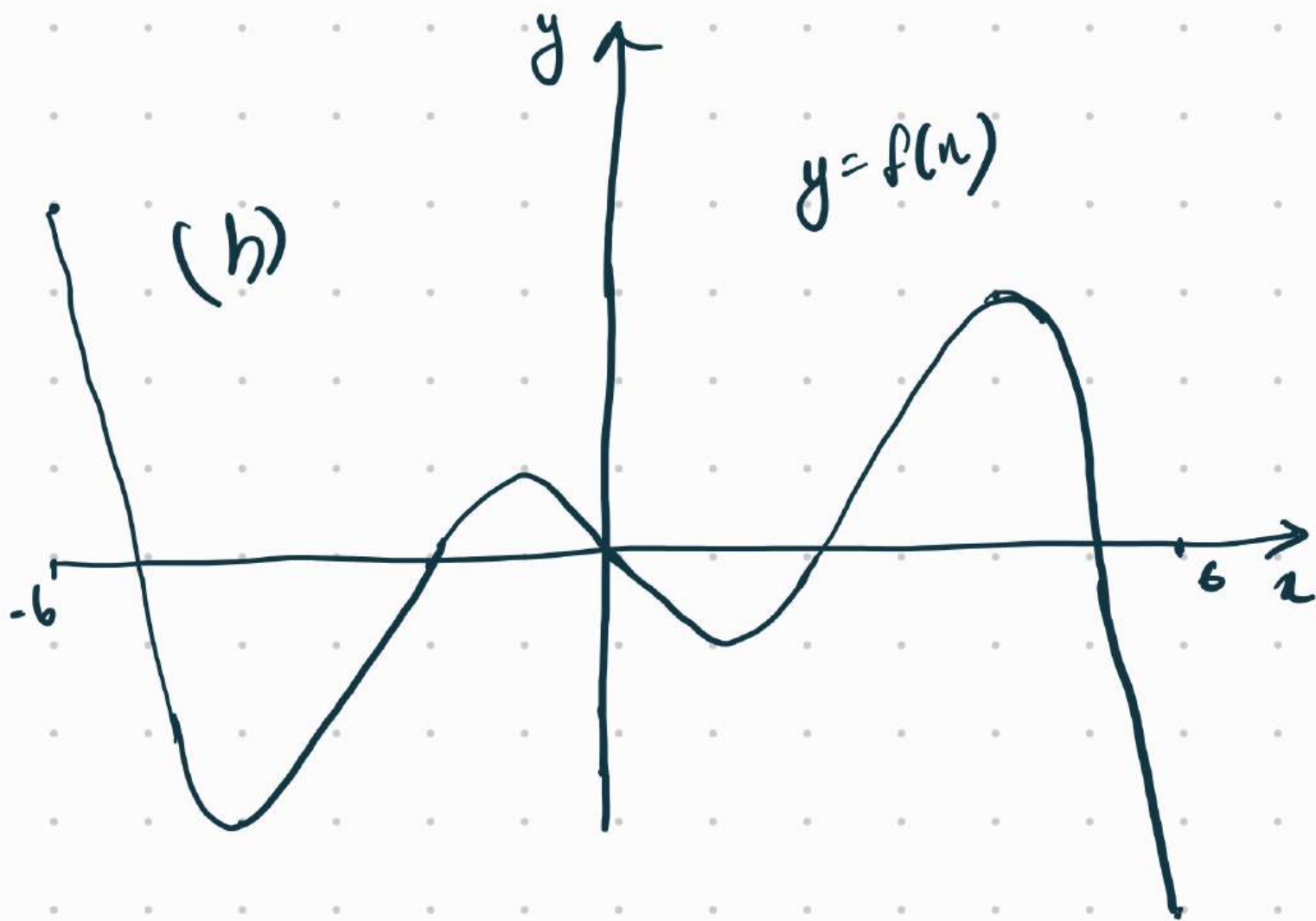
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9.

(a)

$$y = |f(x)|$$





(c) Given $\int_0^4 f(|x|) dx = 1.6$

(i) $\int_{-4}^0 f(x) dx = -1.6$ (odd f^n)

(ii) $\int_{-4}^4 f(|x|) + f(x) dx$
 $= \int_{-4}^4 f(|x|) dx + \int_{-4}^4 f(x) dx$

$$= 1.6 \times 2 + 0$$

$$= 3.2$$



odd function.

$$\int_{-4}^0 f(x) dx \quad \int_0^4 f(x) dx$$

will cancel off.

10. $f(x) = \frac{4x+2}{x-2}, x \neq 2.$

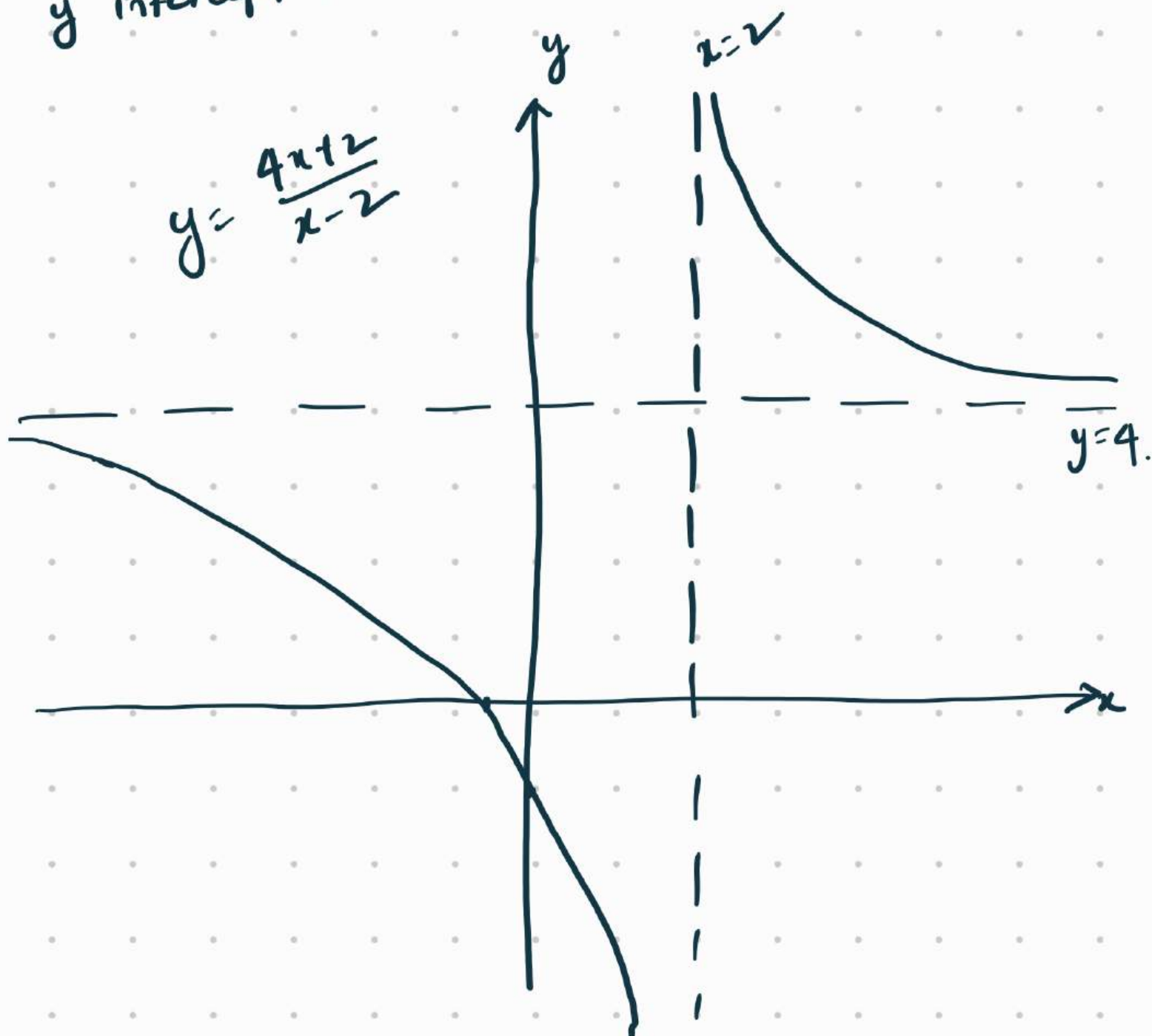
(a) $= 4 + \frac{10}{x-2}$

vertical asymptote: $x=2.$

Horizontal asymptote: $y=4.$

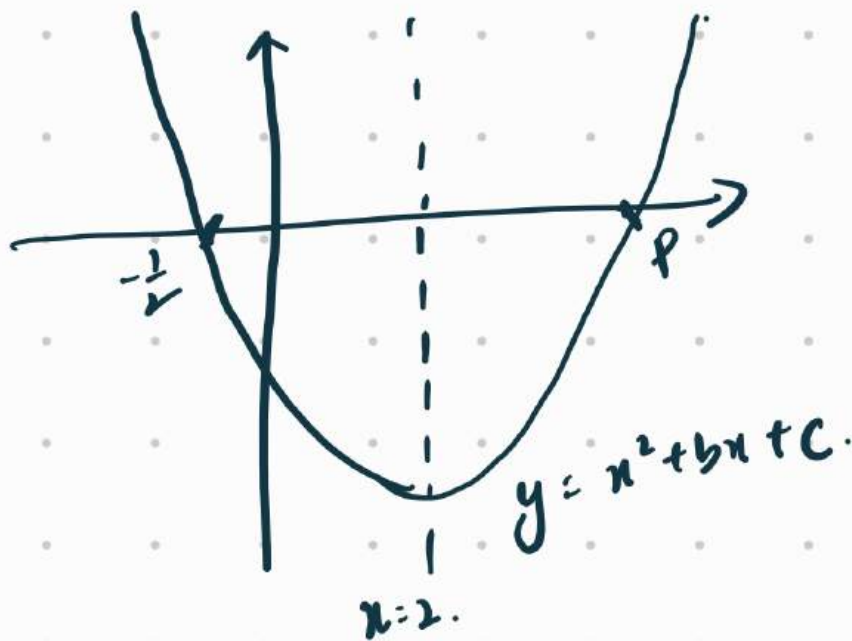
x intercept: when $y=0, x=-\frac{1}{2}$

y intercept: when $x=0, y=-1$



(b) Range of f .
 $y \in \mathbb{R}, y \neq 4$.

(c) $g(x) = x^2 + bx + c$
Axis of symmetry: $x = 2$.



$$\frac{p - \frac{1}{2}}{2} = 2 \Rightarrow p = 4.5 = \frac{9}{2}.$$

(d) $(x + \frac{1}{2})(x - \frac{9}{2}) = 0$

$$x^2 - \frac{9}{2}x + \frac{1}{2}x - \frac{9}{4} = 0$$

$$x^2 - 4x - \frac{9}{4} = 0$$

$$\Rightarrow g(x) = x^2 - 4x - \frac{9}{4}$$

$$b = -4, \quad c = -\frac{9}{4}$$

$$\begin{aligned} (e) \quad g(2) &= 4 - 4(2) - \frac{9}{4} \\ &= -4 - \frac{9}{4} \\ &= -\frac{25}{4} \end{aligned}$$

y-coordinate of vertex = $-\frac{25}{4}$.

$$(f) \quad f(x) = g(x)$$

$$\frac{4x+2}{x-2} = x^2 - 4x - \frac{9}{4}$$

$$4x+2 = (x^2 - 4x - \frac{9}{4})(x-2)$$

$$\Rightarrow x^3 - 2x^2 - 4x^2 + 8x - \frac{9}{4}x + \frac{9}{2} - 4x - 2 = 0$$

$$x^3 - 6x^2 + \frac{7}{4}x + \frac{5}{2} = 0$$

$$\text{product of roots} = \frac{-\frac{5}{2}}{1}$$

$$= -\frac{5}{2}$$

$$11. \quad P(x) = 3x^3 + 5x^2 + x - 1$$

$$(a) \quad P(-1) = 3(-1) + 5(1) - 1 - 1 \\ = 0$$

Hence, $x+1$ is a factor of $P(x)$

(b)

$$P(x) = (x+1)(3x^2 + 2x - 1) \\ = (x+1)(3x-1)(x+1) \\ = (3x-1)(x+1)^2$$

$$\begin{array}{r} x+1 \overline{) 3x^3 + 5x^2 + x - 1} \\ \underline{3x^2 + 2x - 1} \\ 2x^2 + x - 1 \\ \underline{2x^2 + 2x} \\ -x - 1 \\ \underline{-x - 1} \\ 0 \end{array}$$

$$Q(x) = (x+1)(2x+1)$$

$$(c) \quad \frac{1}{Q(x)} = \frac{1}{(x+1)(2x+1)}$$

$$= \frac{A}{x+1} + \frac{B}{2x+1}$$

$$\Rightarrow A(2x+1) + B(x+1) = 1$$

$$2A + B = 0 \quad \text{--- (1)}$$

$$A + B = 1 \quad \text{--- (2)}$$

$$(1) - (2): \quad A = -1$$

$$B = 2.$$

$$\frac{1}{(x+1)(2x+1)} = -\frac{1}{x+1} + \frac{2}{2x+1}$$

$$(d) \frac{1}{(x+1)Q(x)} = \frac{1}{(x+1)^2(2x+1)}$$

$$= \left(\frac{1}{x+1} \right) \left[-\frac{1}{x+1} + \frac{2}{2x+1} \right]$$

$$= -\frac{1}{(x+1)^2} + \frac{2}{(x+1)(2x+1)}$$

$$= -\frac{1}{(x+1)^2} + 2 \left[-\frac{1}{x+1} + \frac{2}{2x+1} \right]$$

$$= \frac{4}{2x+1} - \frac{2}{x+1} - \frac{1}{(x+1)^2}$$

$$(e) \int \frac{1}{(x+1)^2(2x+1)} dx$$

$$= \int \frac{4}{2x+1} - \frac{2}{x+1} - \frac{1}{(x+1)^2} dx$$

$$= 2 \ln|2x+1| - 2 \ln|x+1| + \frac{1}{x+1} + C$$

$$(f) \quad f(x) = \frac{P(x)}{(x+1)Q(x)}, \quad x \neq -1, x \neq -\frac{1}{2}$$

$$(i) \lim_{x \rightarrow -1} f(x) = \lim_{x \rightarrow -1} \left[\frac{(x+1)^2(3x-1)}{(x+1)^2(2x+1)} \right]$$

$$= \lim_{x \rightarrow -1} \left(\frac{3x-1}{2x+1} \right)$$

$$= \frac{-4}{-1} = 4$$

$$(ii) \lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} \left(\frac{3x-1}{2x+1} \right)$$

$$= \lim_{x \rightarrow \infty} \left(\frac{3 - \frac{1}{x}}{2 + \frac{1}{x}} \right)$$

$$= \frac{3}{2}$$

$$12. \quad \phi = (a+bi)^3, \quad a, b \in \mathbb{R}.$$

$$\begin{aligned} (a) \quad (a+bi)^3 &= (a+bi)(a+bi)(a+bi) \\ &= (a^2 + 2abi - b^2)(a+bi) \\ &= a^3 + a^2bi + 2a^2bi - 2ab^2 - b^2a - b^3i \\ &= (a^3 - 3ab^2) + (3a^2b - b^3)i \end{aligned}$$

$$(i) \quad \operatorname{Re}(\phi) = a^3 - 3ab^2$$

$$(ii) \quad \operatorname{Im}(\phi) = 3a^2b - b^3.$$

$$(b) \quad (1 + \sqrt{3}i)^3$$

$$a=1, b=\sqrt{3}$$

$$= [1 - 3(3)] + [3\sqrt{3} - (\sqrt{3})^3]i$$

$$= -8 + (3\sqrt{3} - 3\sqrt{3})i$$

$$= -8.$$

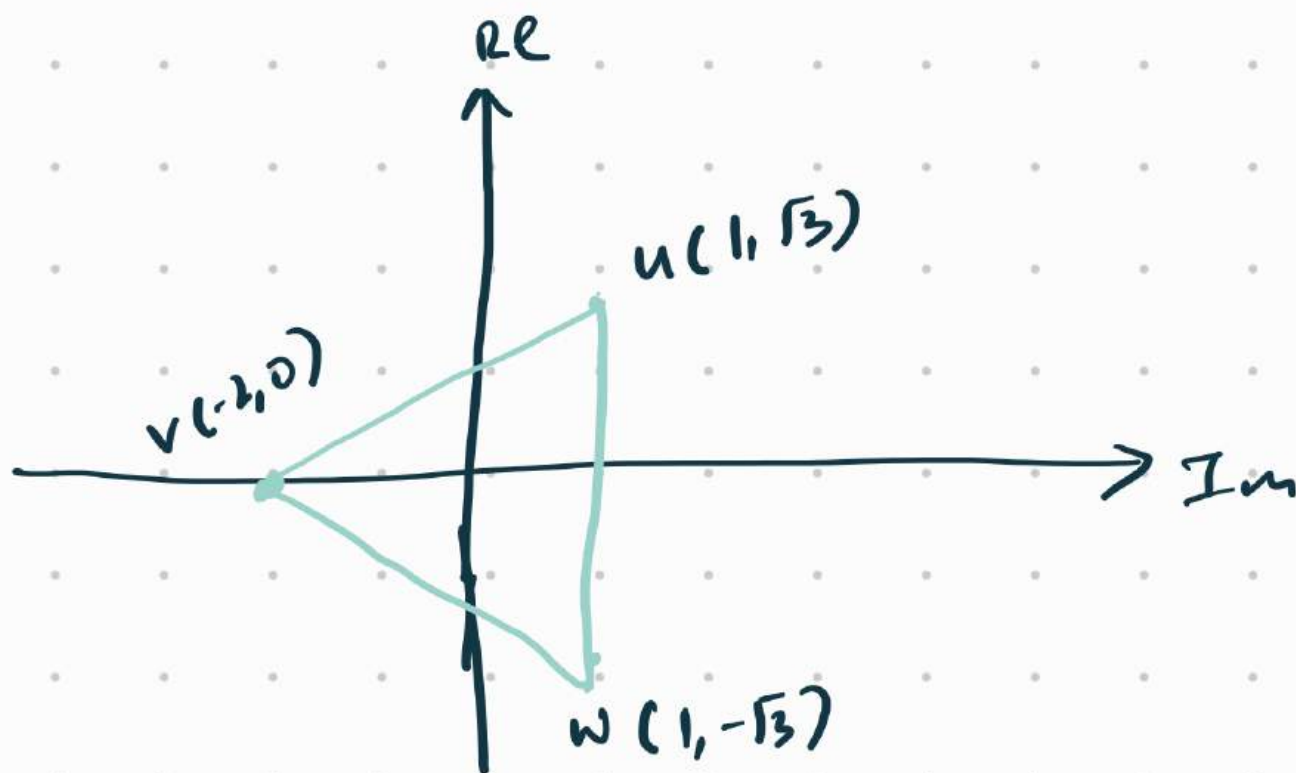
$$z^3 = -8$$

$$u = 1 + \sqrt{3}i \quad v \in \mathbb{R}$$

(c) $v = -2$

$$w = 1 - \sqrt{3}i \quad (\text{conjugate})$$

(d)



$$\begin{aligned} \text{Area of } \triangle uvw &= \frac{1}{2} (2\sqrt{3}) (3) \\ &= 3\sqrt{3} \text{ units}^2 \end{aligned}$$

(e) $u = 1 + \sqrt{3}i = 2e^{i(\frac{\pi}{3})}$

$$v = -2 = 2e^{i\pi}$$

$$w = 1 - \sqrt{3}i = 2e^{i(-\frac{\pi}{3})}$$

$$u' = 2e^{i\frac{\pi}{3}} \cdot e^{i\frac{\pi}{4}} = 2e^{i\frac{7\pi}{12}}$$

$$v' = 2e^{i\pi} \cdot e^{i\frac{\pi}{4}} = 2e^{i\frac{5\pi}{4}} \\ = 2e^{i(-\frac{3\pi}{4})}$$

$$w' = 2e^{i(-\frac{\pi}{3})} e^{i\frac{\pi}{4}} = 2e^{i(-\frac{\pi}{12})}$$

(f) u', v', w' are solutions of
 $z^3 = c + di$

$$(u')^3 = (2e^{i\frac{7\pi}{12}})^3 = 8e^{i\frac{7\pi}{4}} \\ = 8e^{i(-\frac{\pi}{4})} \\ = 8[\cos(-\frac{\pi}{4}) + i\sin(-\frac{\pi}{4})] \\ = 8\left(\frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2}\right) \\ = 4\sqrt{2} - i(4\sqrt{2})$$

$$c = 4\sqrt{2}, \quad d = -4\sqrt{2}.$$

$$(9) \quad z^n = \alpha$$

arguments of u, v, w, u', v', w'

$$\frac{\pi}{3}, \pi, -\frac{\pi}{3}, \frac{7\pi}{12}, -\frac{5\pi}{4}, -\frac{\pi}{12}.$$

$$\Rightarrow \frac{4\pi}{12}, \frac{12\pi}{12}, -\frac{4\pi}{12}, \frac{7\pi}{12}, -\frac{9\pi}{12}, -\frac{\pi}{12}.$$

arranging in order,

$$-\frac{9\pi}{12}, -\frac{4\pi}{12}, -\frac{\pi}{12}, \frac{4\pi}{12}, \frac{7\pi}{12}, \frac{12\pi}{12}.$$

arguments differ by $\frac{3\pi}{12}$ and $\frac{5\pi}{12}$.

$$n=12 \dots$$

